

L02 — Algorithm Complexity: Time and Space; Time-Space Tradeoff

Unit: 1 | Chapter: Fundamentals of Data Structures & Arrays | BT Level: BT4 | CO: CO1

Learning Objectives

- Explain what algorithm complexity means and why it matters
 - Differentiate between time complexity and space complexity
 - Analyze simple algorithms for their time and space usage
 - Understand and apply the time-space tradeoff concept
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1. What is Algorithm Complexity?

Algorithm complexity measures the resources (time and memory) an algorithm consumes relative to the size of its input, denoted n .

We study complexity because:

- Hardware is finite — we cannot run an $O(2^n)$ algorithm on $n = 100$
 - Scalability matters — an algorithm working for $n = 1000$ may be unusable for $n = 10^6$
 - Complexity helps us compare algorithms **independent of hardware**
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2. Time Complexity

Time complexity counts the number of basic operations (comparisons, assignments, arithmetic) an algorithm performs as a function of input size n .

2.1 Types of Analysis

Type	Definition	Example
Best Case	Minimum operations (most favorable input)	Linear search: element is at index 0 $\rightarrow O(1)$
Worst Case	Maximum operations (least favorable input)	Linear search: element not found $\rightarrow O(n)$
Average Case	Expected operations over all inputs	Linear search: element at middle $\rightarrow O(n/2) = O(n)$

In practice, we focus on **worst-case analysis** — it gives an upper bound guarantee.

2.2 Counting Operations Example

```
# Find maximum in an array
```

```
def find_max(arr):
    max_val = arr[0]          # 1 operation
    for i in range(1, n):     # n-1 iterations
        if arr[i] > max_val:  # 1 comparison per iteration
            max_val = arr[i]  # at most 1 assignment
    return max_val            # 1 operation
```

Total operations $\approx 1 + 3(n-1) + 1 = 3n - 1 \rightarrow O(n)$

3. Space Complexity

Space complexity measures the amount of memory an algorithm uses relative to input size n .

It includes:

- **Input space:** Memory for storing the input
- **Auxiliary space:** Extra memory used by the algorithm (variables, stack frames, temporary arrays)

Usually we care about **auxiliary space** (extra space beyond input).

3.1 Space Complexity Examples

$O(1)$ auxiliary space – only a few variables

```
def sum_array(arr, n):
    total = 0
    for x in arr:
        total += x
    return total
```

$O(n)$ auxiliary space – stores a copy of input

```
def reverse_array(arr, n):
    result = [0] * n          #  $O(n)$  space
    for i in range(n):
        result[n-1-i] = arr[i]
    return result
```

4. Common Complexity Classes

Notation	Name	Example
$O(1)$	Constant	Array access by index
$O(\log n)$	Logarithmic	Binary search
$O(n)$	Linear	Linear search
$O(n \log n)$	Linearithmic	Merge sort, heap sort
$O(n^2)$	Quadratic	Bubble sort, insertion sort
$O(n^3)$	Cubic	Naive matrix multiplication
$O(2^n)$	Exponential	Subset enumeration

$O(n!)$ Factorial Brute-force TSP

Growth Rate Comparison (for $n = 1000$):

Complexity	Operations
$O(1)$	1
$O(\log n)$	~ 10
$O(n)$	1,000
$O(n \log n)$	$\sim 10,000$
$O(n^2)$	1,000,000
$O(2^n)$	$\sim 10^{301}$ (impossible)

5. Time-Space Tradeoff

A **time-space tradeoff** is a situation where using more memory allows an algorithm to run faster, or using less memory forces it to run slower.

Core idea: You can often trade space for time or time for space depending on your system's constraints.

5.1 Examples of Time-Space Tradeoff

Example 1: Lookup Table (Memoization)

```
# Without memoization -  $O(2^n)$  time,  $O(n)$  space (call stack)
def fib_slow(n):
    if n <= 1: return n
    return fib_slow(n-1) + fib_slow(n-2)

# With memoization -  $O(n)$  time,  $O(n)$  space
memo = {}
def fib_fast(n):
    if n in memo: return memo[n]
    if n <= 1: return n
    memo[n] = fib_fast(n-1) + fib_fast(n-2)
    return memo[n]
```

By storing previously computed results (extra $O(n)$ space), we reduce time from $O(2^n)$ to $O(n)$.

Example 2: Precomputed Prefix Sums

```
# Without prefix sums:  $O(n)$  per query
def range_sum_slow(arr, l, r):
    return sum(arr[l:r+1]) #  $O(r-l+1)$  per query

# With prefix sums:  $O(n)$  precomputation,  $O(1)$  per query
def precompute(arr, n):
    prefix = [0] * (n+1)
```

```

for i in range(n):
    prefix[i+1] = prefix[i] + arr[i]
return prefix

def range_sum_fast(prefix, l, r):
    return prefix[r+1] - prefix[l]    # O(1)

```

Example 3: Hash Table vs Binary Search

Approach	Space	Search Time
Binary Search on sorted array	O(1) extra	O(log n)
Hash Table	O(n)	O(1) average

6. Factors Affecting Time Complexity

1. **Algorithm design** — the dominant factor
2. **Data structures used** — linked list vs array vs hash table
3. **Input size and characteristics** — sorted vs unsorted, dense vs sparse
4. **Platform** — interpreted vs compiled language (constant factor, not asymptotic)

7. Practical Guidelines

- Prefer **$O(n \log n)$** over **$O(n^2)$** for $n > 10^4$
- If space is unlimited: use memoization/caching to speed up repetitive computation
- If memory is limited: accept slower algorithms or streaming approaches
- Profile before optimizing — find the real bottleneck

Summary

- **Time complexity** measures operations as a function of n ; analyzed in best, worst, and average cases.
- **Space complexity** measures memory usage (focus on auxiliary space).
- Common classes: $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n)$.
- **Time-space tradeoff**: Trading extra memory for faster execution (memoization, lookup tables, prefix sums).

References

- Cormen et al., *Introduction to Algorithms*, Ch. 3 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 2 (Springer, 2020)
- Laaksonen, *Guide to Competitive Programming*, Ch. 2 (Springer, 2024)